

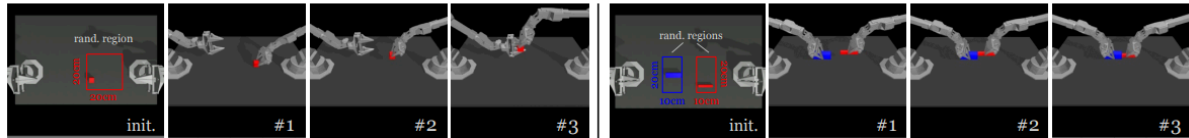
Task 1 Report (Part B)

For Part B, I selected the fine-tuned SmolVLA checkpoint trained on the lerobot/aloha_sim_insertion_human dataset.

Fine-tuned SmolVLA model:

https://huggingface.co/c27e/smolvla_aloha_sim_insertion_human

Dataset: https://huggingface.co/datasets/lerobot/aloha_sim_insertion_human



Left: Cube Transfer. Transfer the red cube to the other arm. The right arm touches (#1) and grasps (#2) the red cube, then hands it to the left arm.
Right: Bimanual Insertion. Insert the red peg into the blue socket. Both arms grasp (#1), let socket and peg make contact (#2) and insertion.

The model was trained on the ALOHA bimanual robot system, which features two 7-DOF collaborative arms for coordinated manipulation tasks.

I saw different fine-tuned models on Hugging Face, and most of them were for grabbing single objects. I wanted to combine two arms in one action; that is why I chose this model, which was more complex than the others. As you can see from the image above, the model was trained to perform one specific task, which was inserting the peg into the socket. The important thing here is that the training environment and robotic arm must match at inference time; otherwise, the movements of the robotic arms are not meaningful. I tested with different prompts to observe the model. For example, when I say “pick up the objects and let them fall,” the arms moved through objects but couldn't even pick them up. When I checked the specific prompt used during training, it was “Insert the peg into the socket” (it can be found in the tasks.parquet metadata file).

Dataset / Policy Input-Output Format

From the checkpoint configuration, the expected data format is:

- Visual input: `observation.images.top` with shape `[3, 480, 640]`
- State input: `observation.state` with shape `[14]`
- Action output: `action` with shape `[14]`

Since there were two robotic arms with 7 DOF, there are 14 outputs.

The rollout format is also defined by the checkpoint:

- `chunk\_size = 50`
- `n\_action\_steps = 50`

So, instead of predicting one action per step, the policy predicts a 50-step action sequence per query.

This setup was used to stay fully consistent with how the selected checkpoint was trained

1. Same task family (ALOHA insertion).
2. Same observation schema (top camera + 14D robot state).
3. Same action schema (14D control output).
4. Same chunked rollout behavior (50-step chunks).

By using the same configuration as in training, the inference results were as expected; the two arms worked in parallel and accurately.